Handshaking Lemma

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University of California, San Diego

Outline

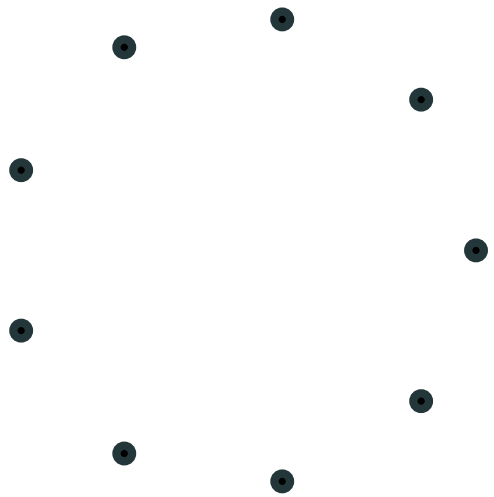
Handshaking Lemma

Total Degree

Revisiting a Puzzle

Puzzle

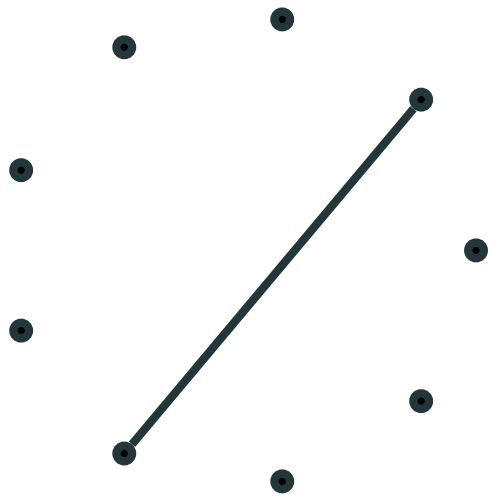
Is it possible to connect some pairs of nine points by segments so that each point is connected to five other points?



Revisiting a Puzzle

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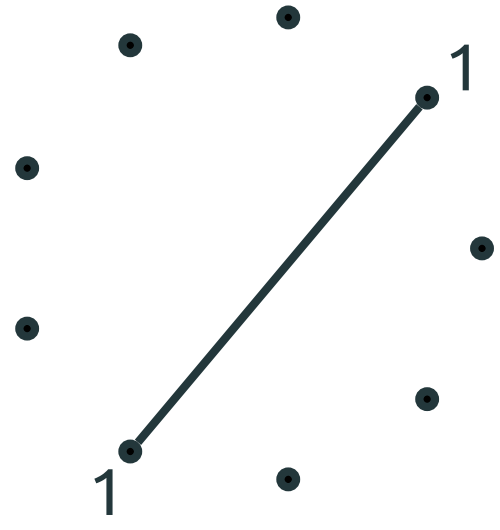
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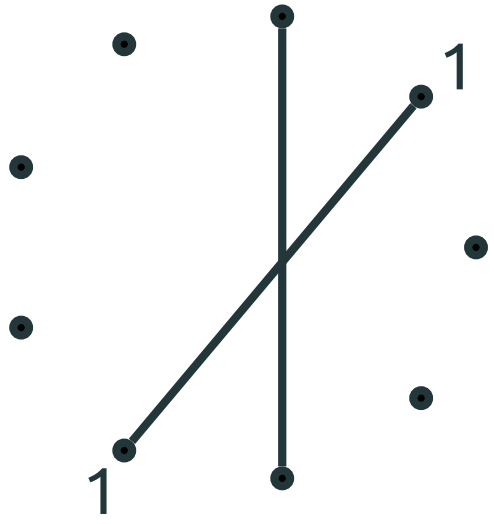
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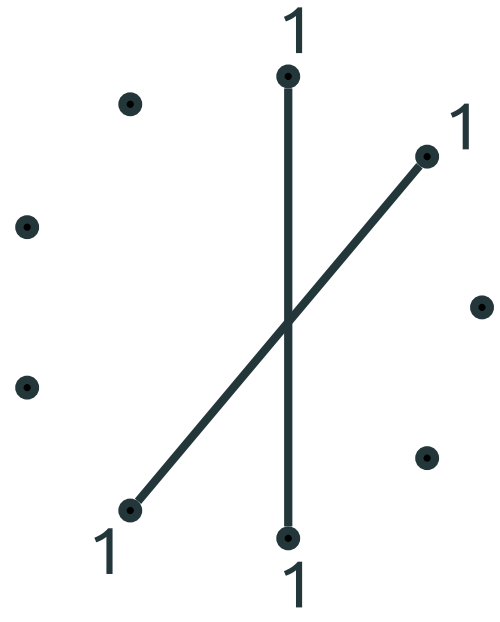
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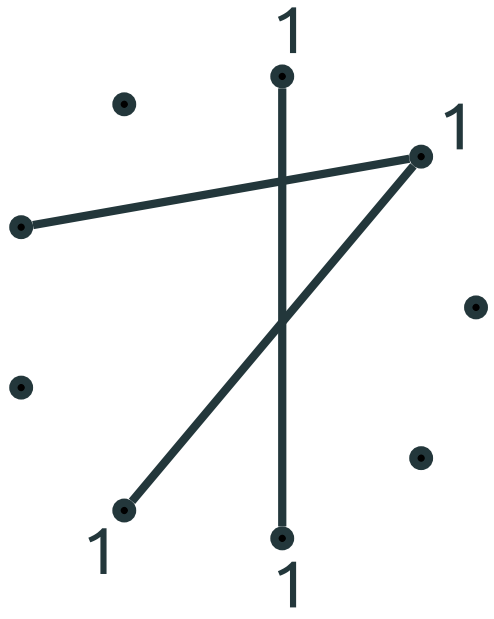
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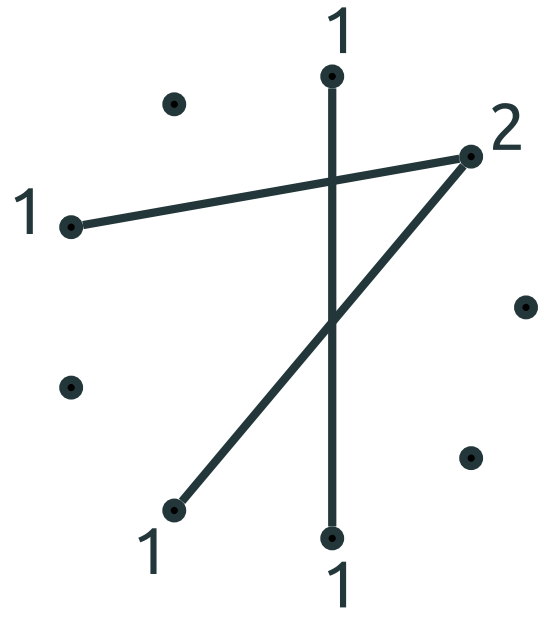
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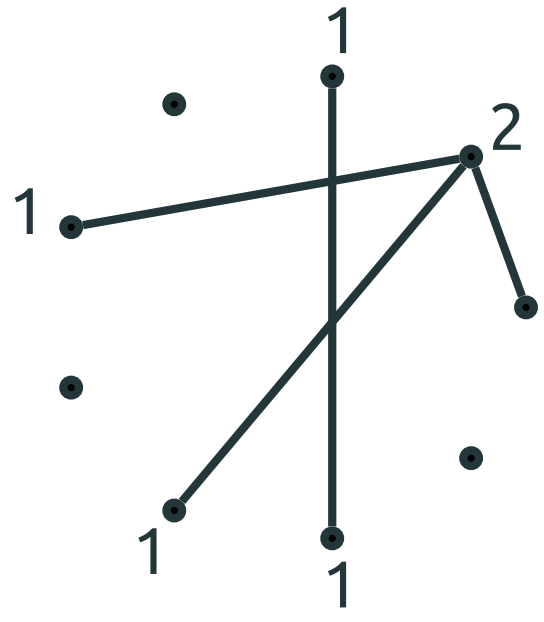
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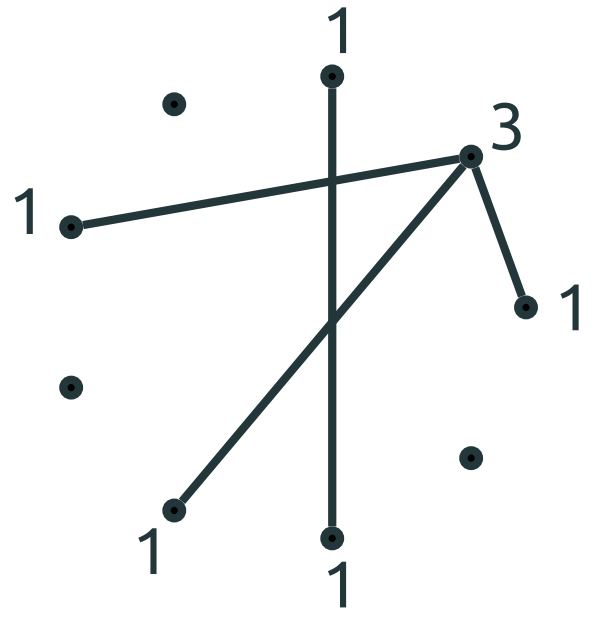
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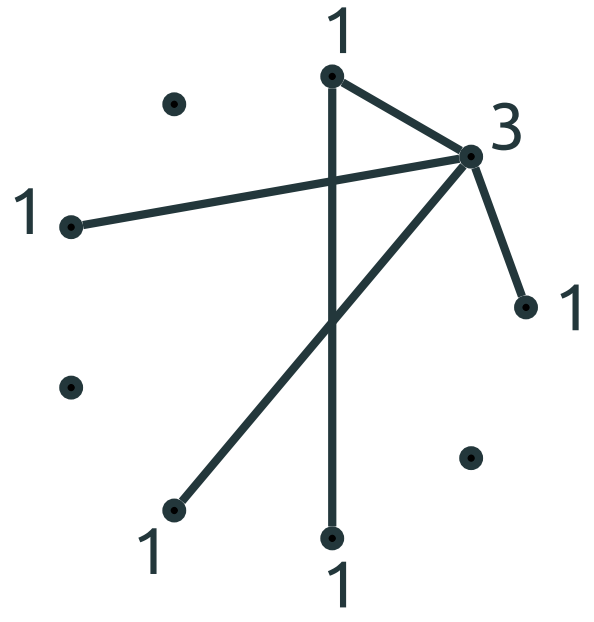
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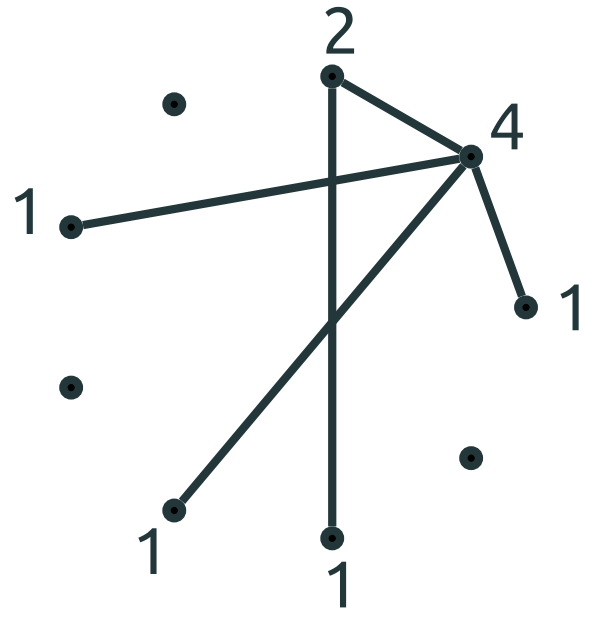
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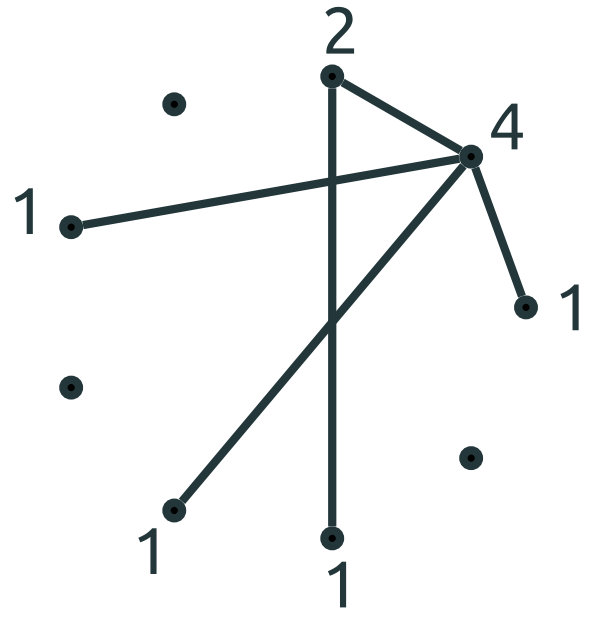
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Revisiting a Puzzle

Puzzle

Is it possible to connect some pairs of nine points by segments so that each point is connected to five other points?



the number of odd points is always even, hence we will never reach a situation when there are 9 points of degree 5

Handshaking



- Before a business meeting, some people shook hands. Then the number of people who made an odd number of handshakes is even

Handshaking



- Before a business meeting, some people shook hands. Then the number of people who made an odd number of handshakes is even
- In graph terms: A graph has an even number of odd nodes

Degree Sum Formula

Lemma

For any graph $G(V, E)$, the sum of degrees of all its nodes is twice the number of edges:

$$\sum_{v \in V} \text{degree}(v) = 2 \cdot |E| .$$

Degree Sum Formula

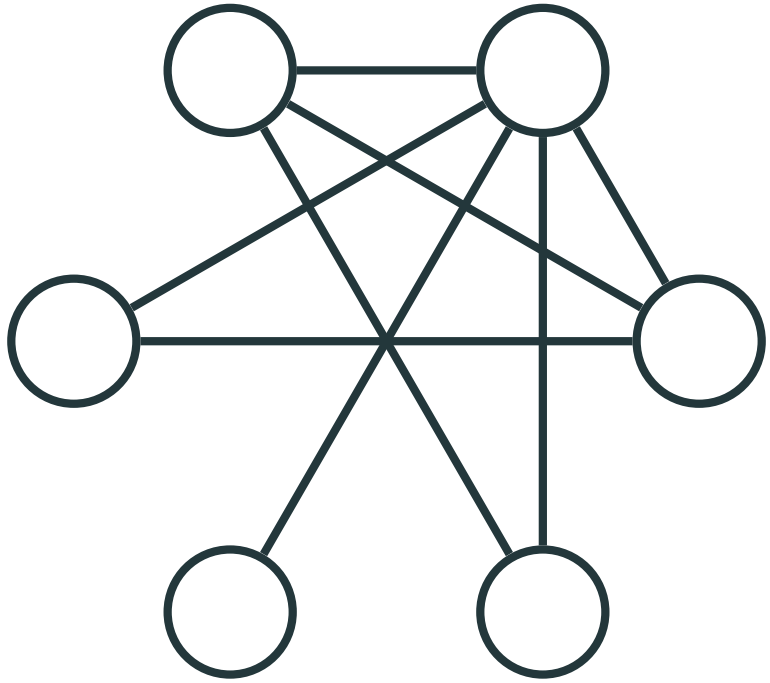
Lemma

For any graph $G(V, E)$, the sum of degrees of all its nodes is twice the number of edges:

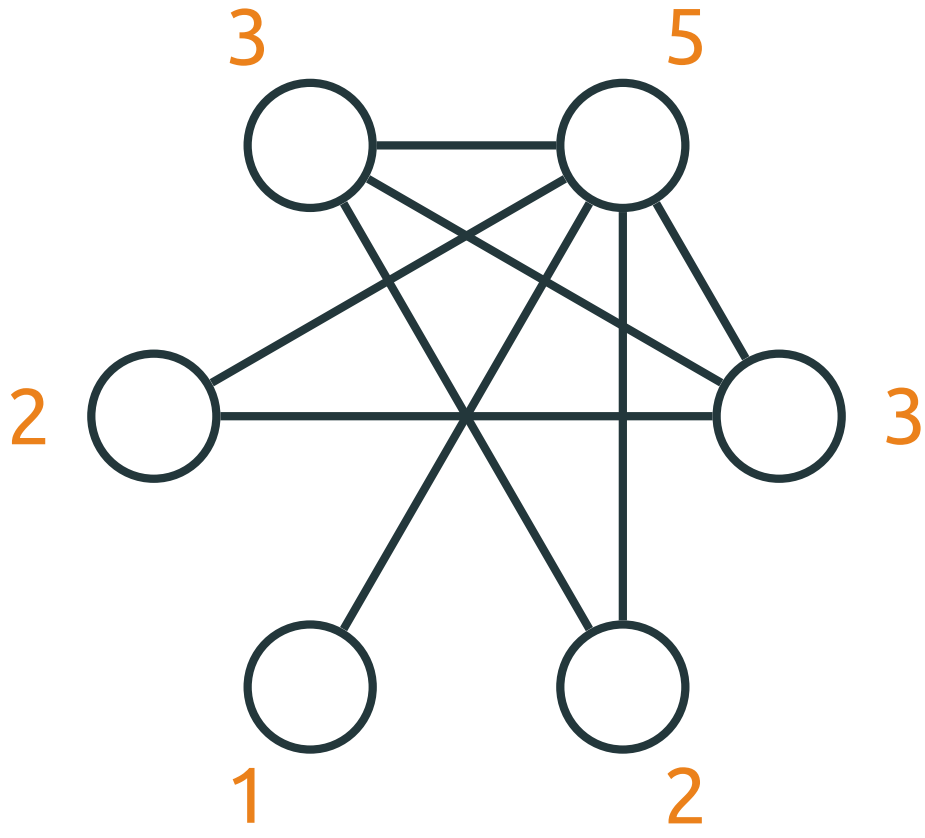
$$\sum_{v \in V} \text{degree}(v) = 2 \cdot |E| .$$

Implies the handshaking lemma: if a graph had an odd number of odd nodes, then the sum of degrees would be also odd.

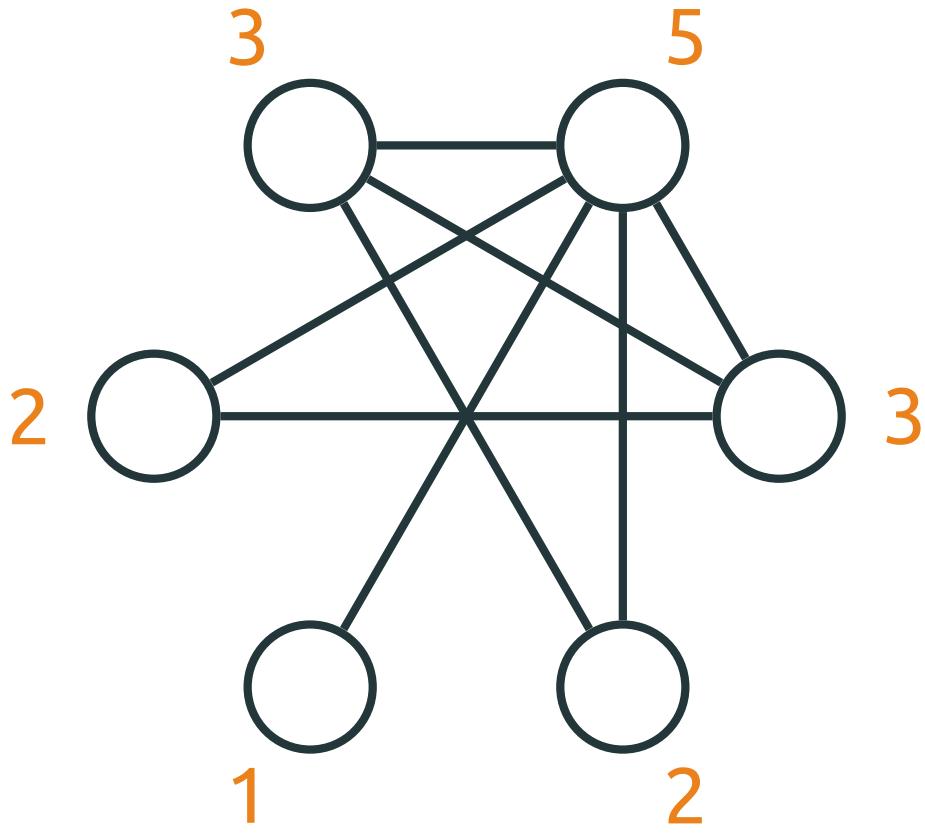
Example



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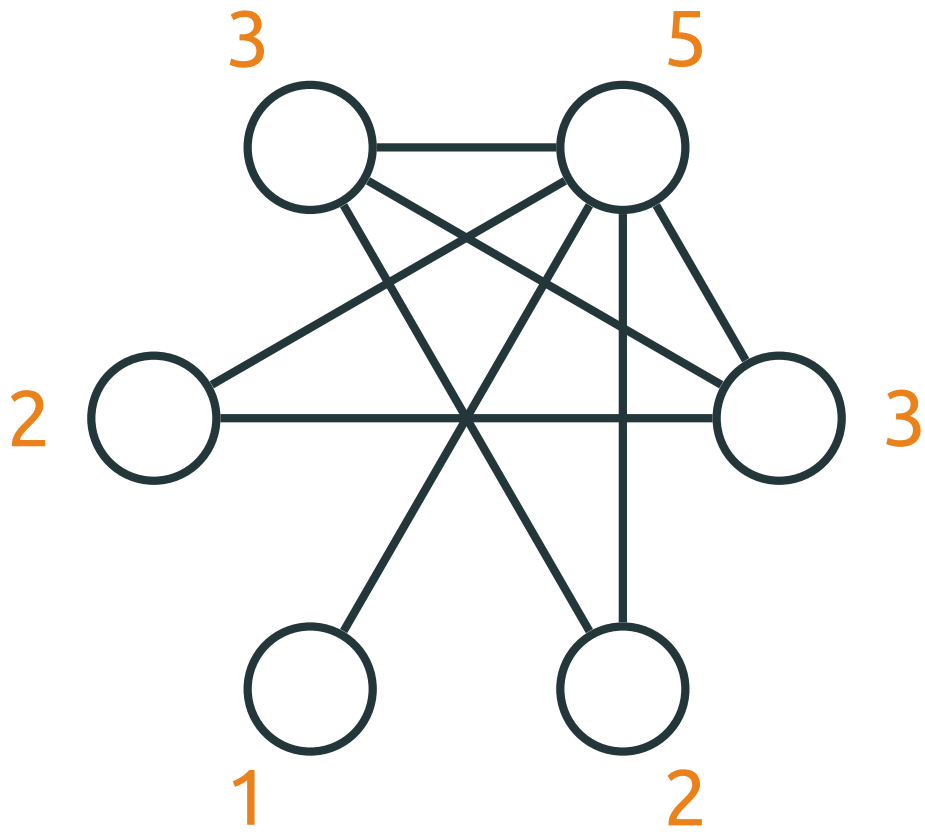


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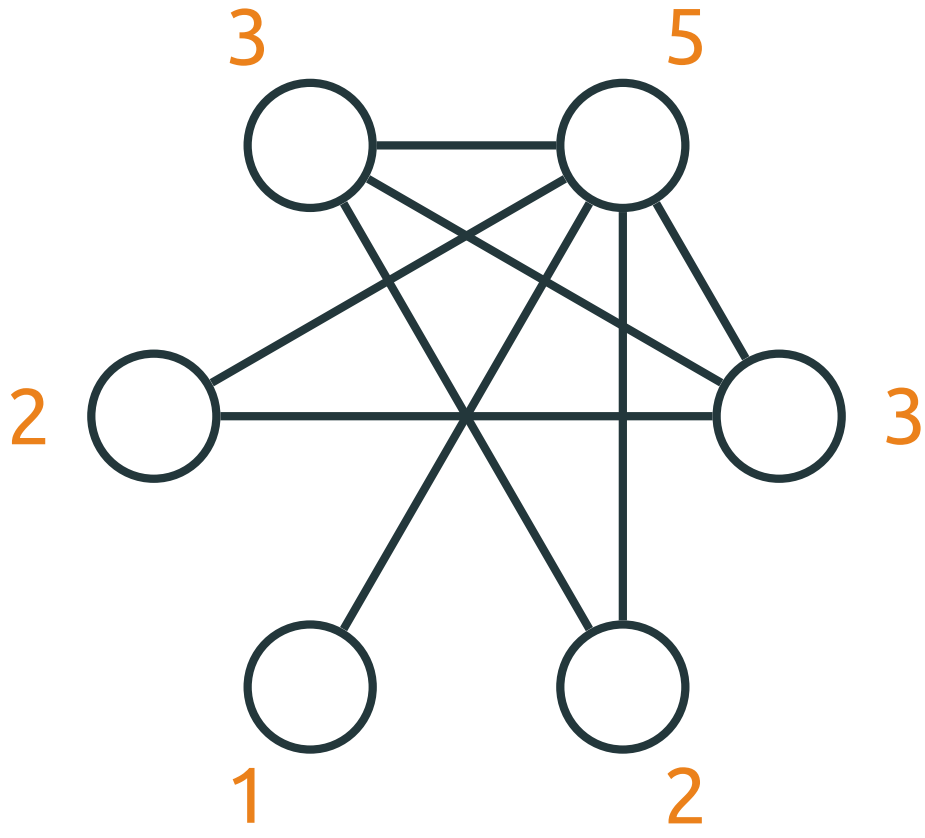


$$3 + 5 + 3 + 2 + 1 + 2 = 2 \cdot 8$$

Proof

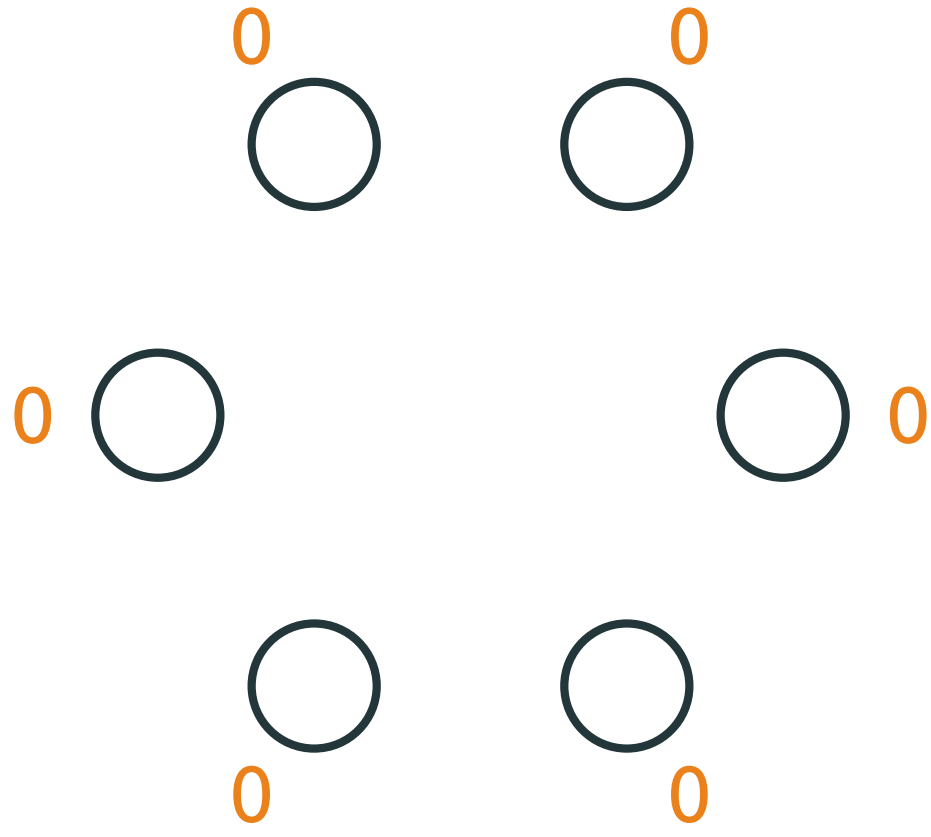


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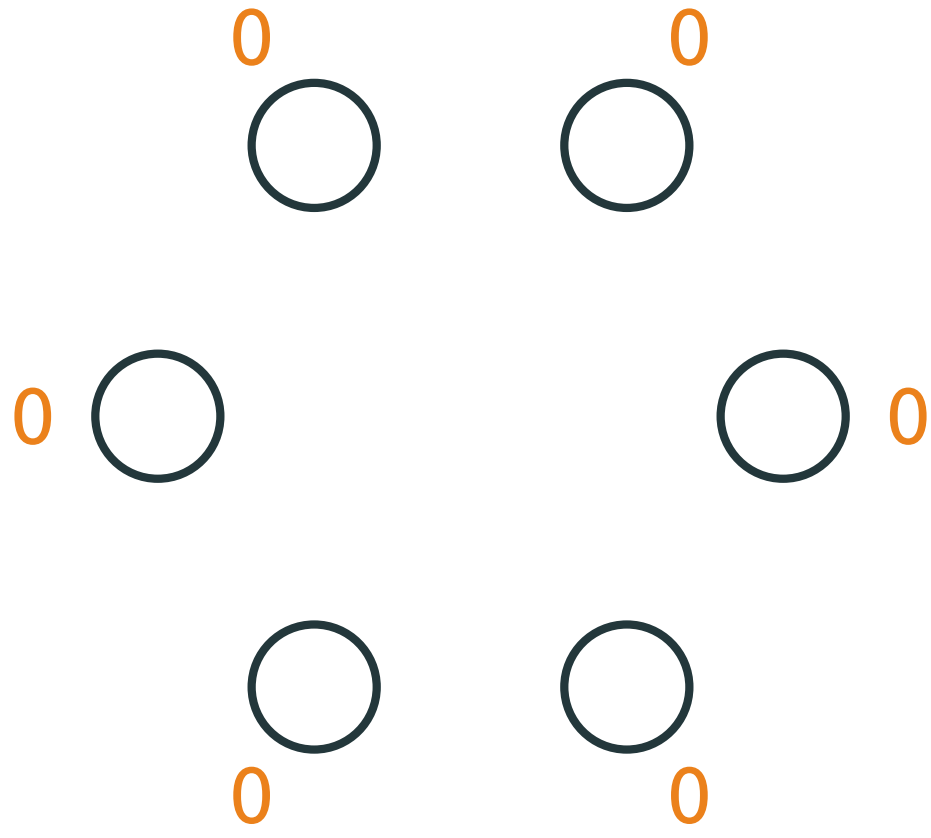
remove all edges

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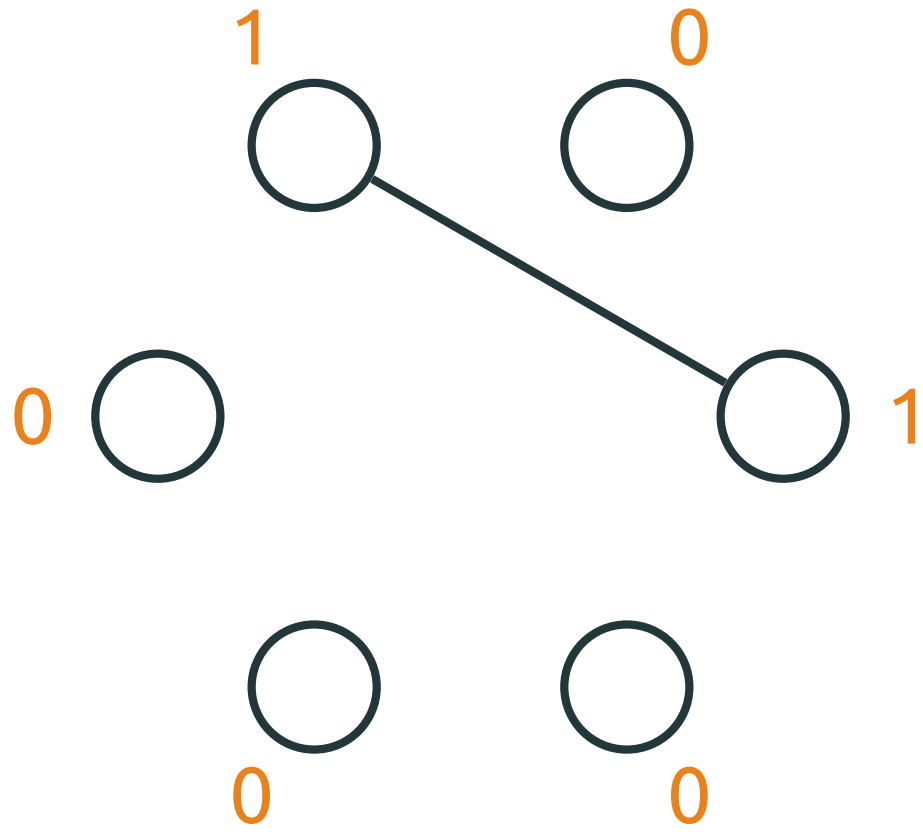
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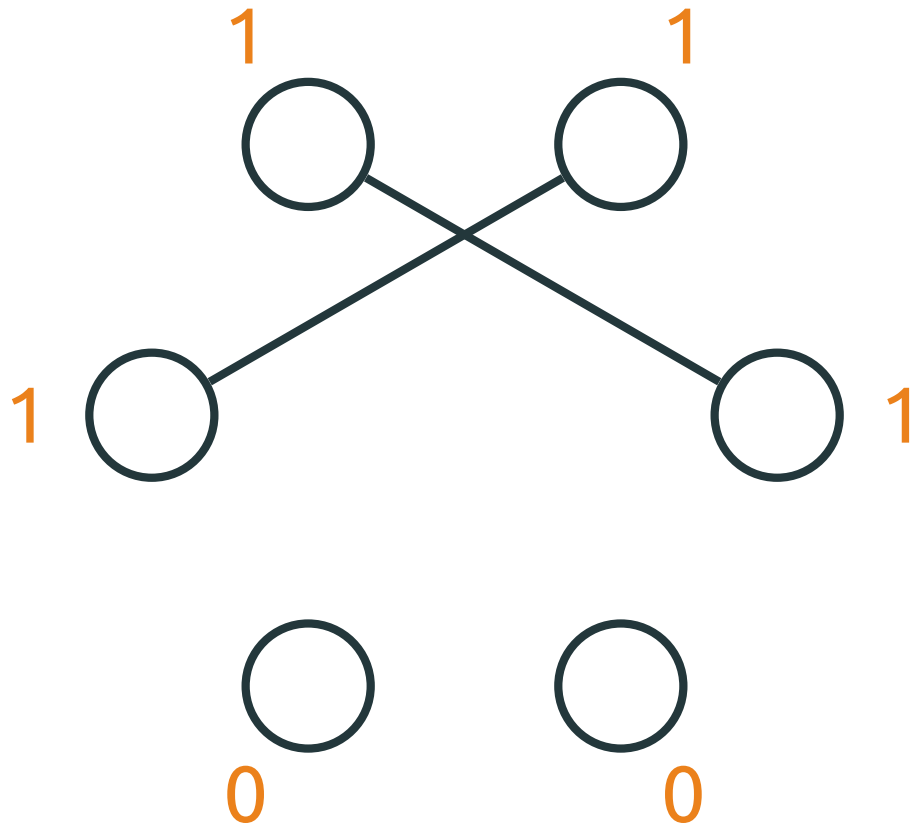
now, start adding them back, one by one

Proof



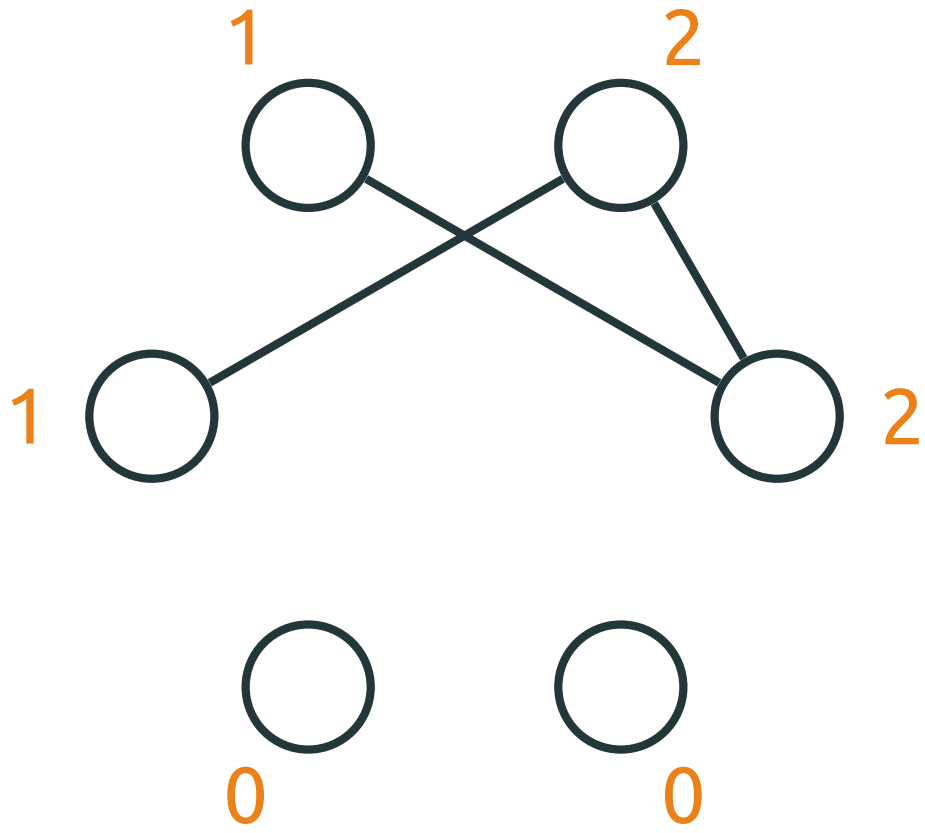
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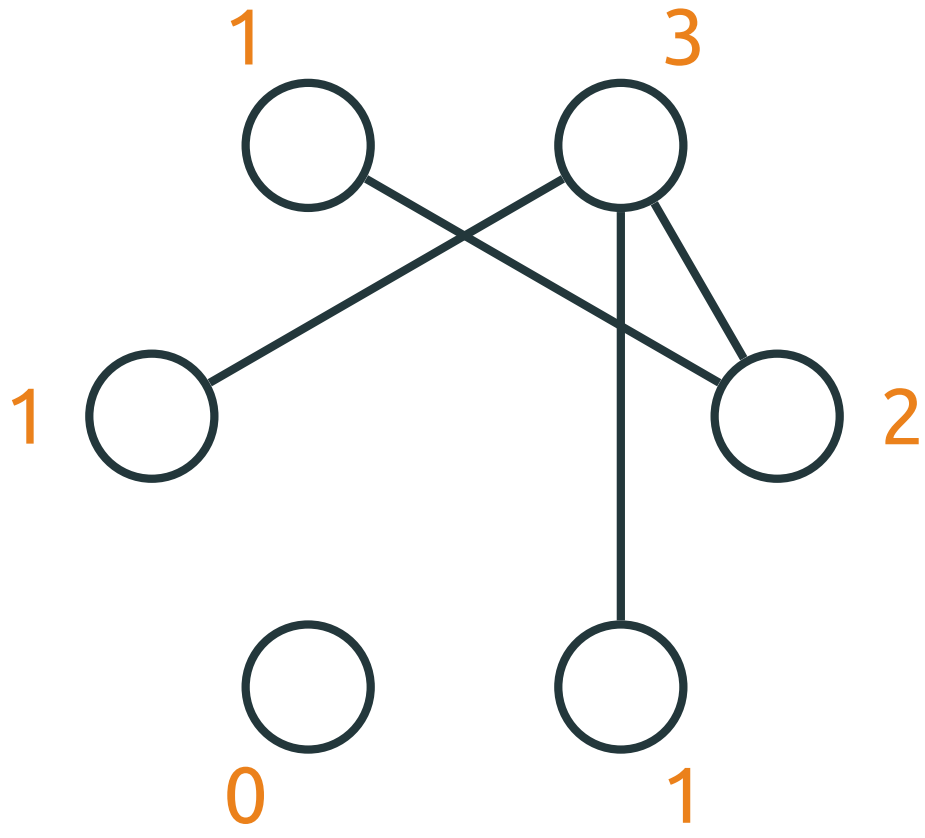
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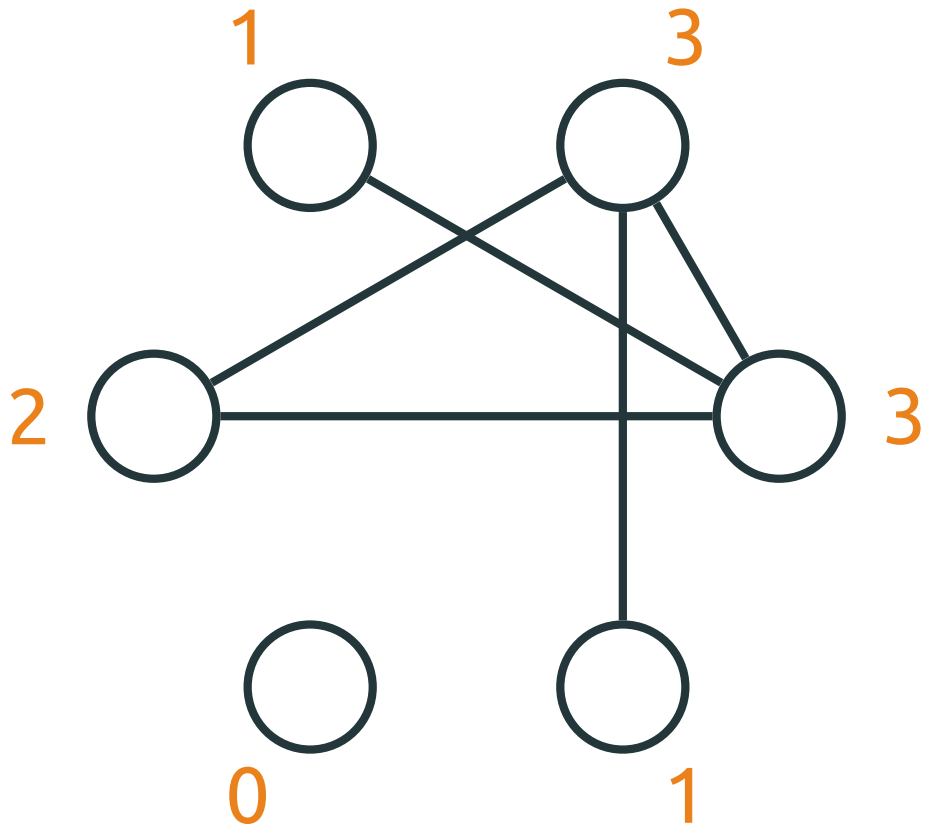
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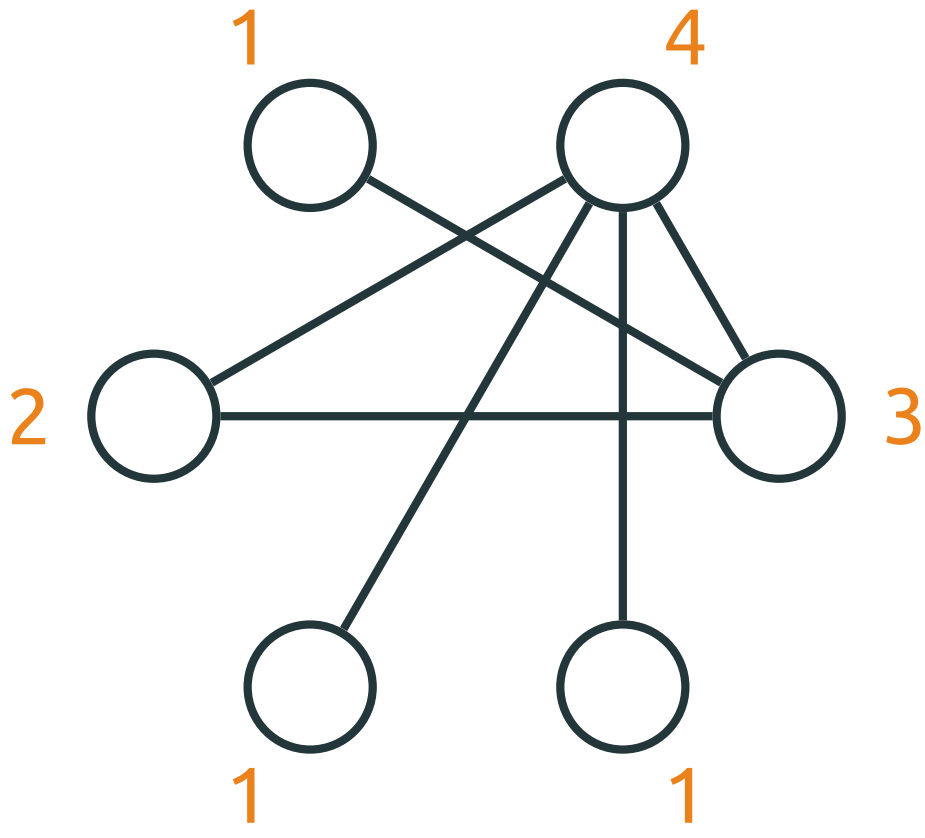
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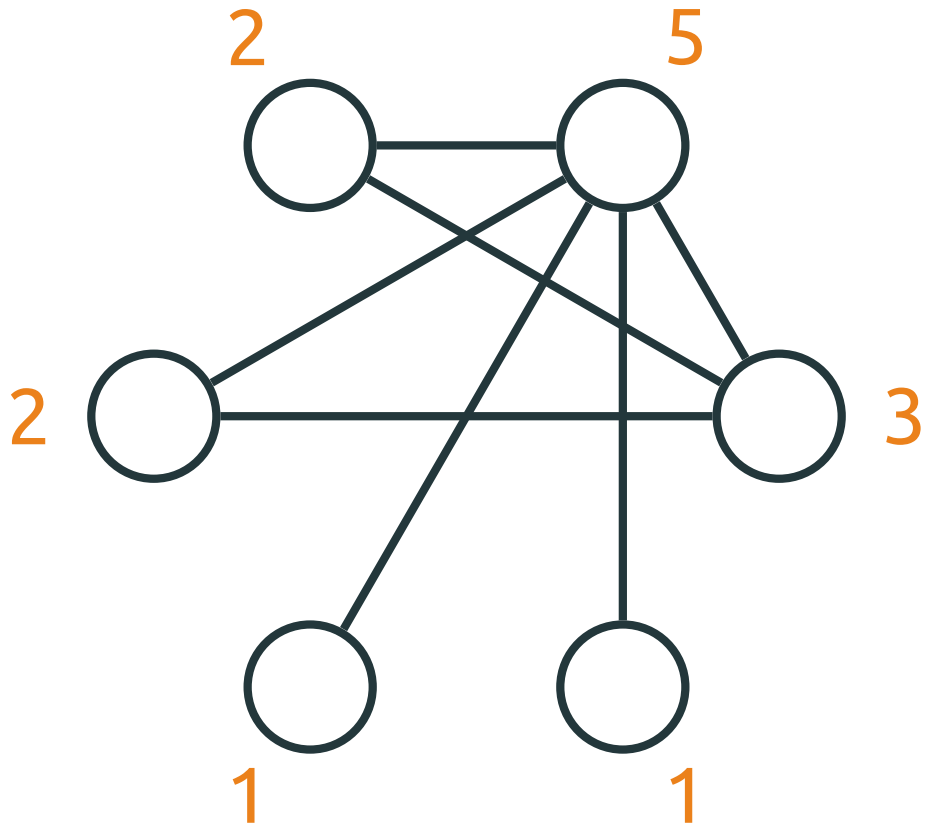
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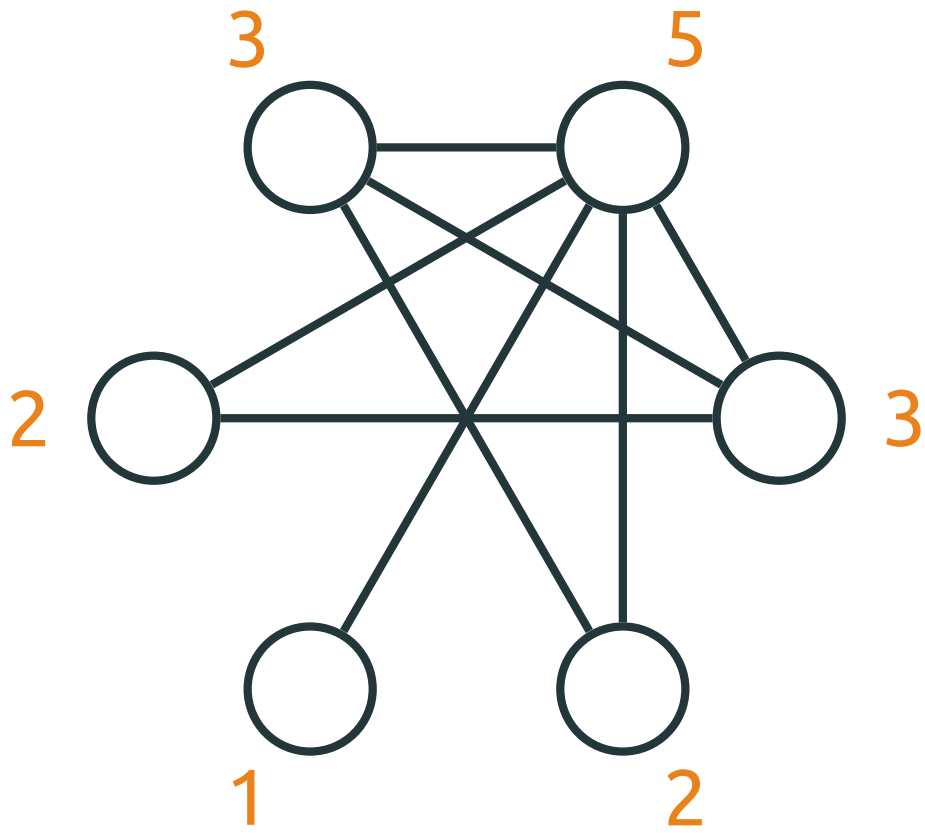
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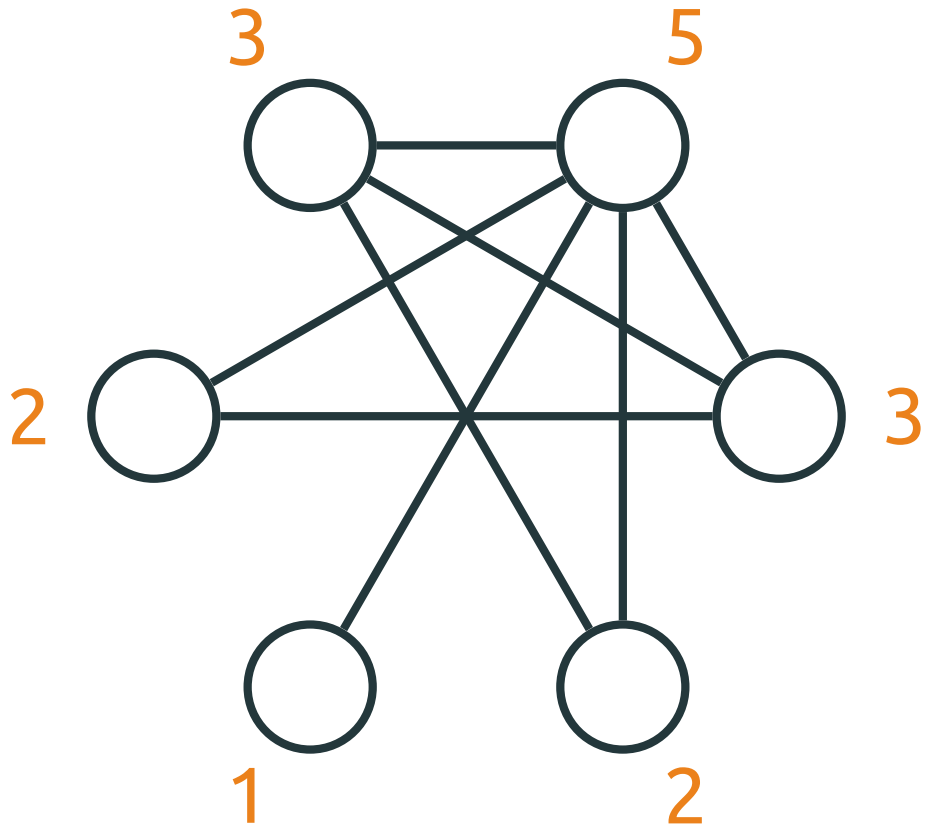
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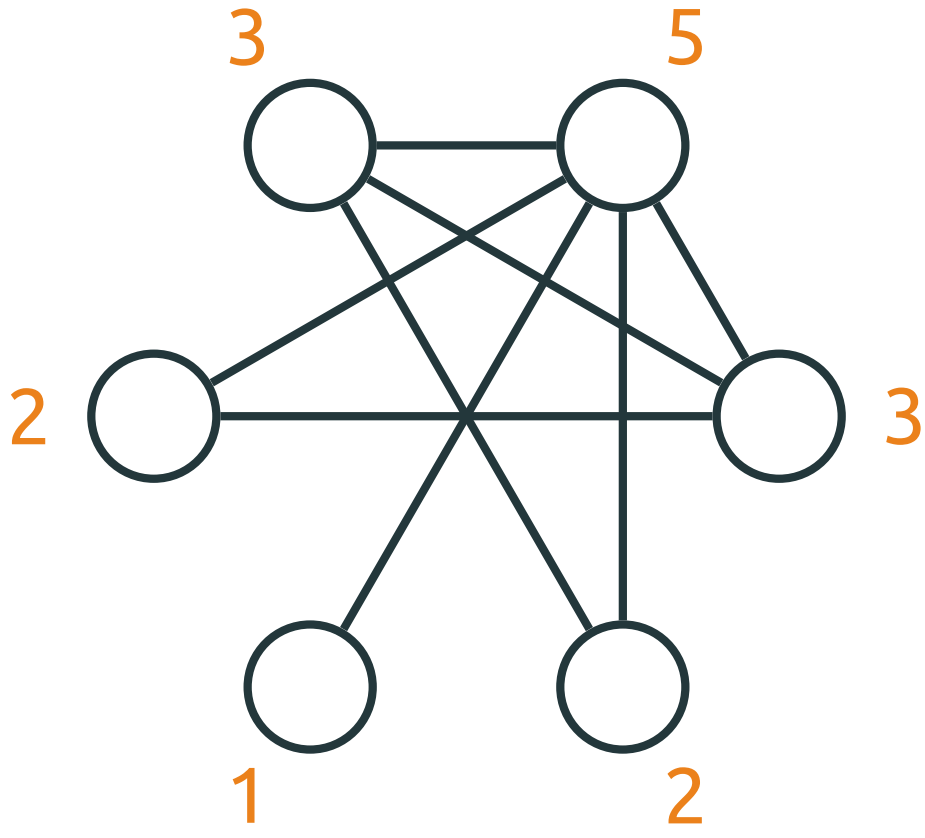
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Proof



each edge contributes 2 to the sum of degrees

Proof



as well as to twice the number of edges!

Summary

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- Base case: the formula holds if there are no edges (all degrees are equal to 0)
- Induction step: when we add an edge, the sum of degrees increases by 2

Outline

Handshaking Lemma

Total Degree

Exam

Problem

At an exam, each of 20 students solved 3 problems. Each problem was solved by 5 students. What was the number of problems?



https://commons.wikimedia.org/wiki/File%3AATC_Admission_Exam.JPG

Solution

- Assume that each student wrote
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- Then, there are $20 \times 3 = 60$ pieces of paper
- For each problem, let's stack together all its five solutions
- Thus, the number of problems is $60/5 = 12$

In Graph Terms

20 students



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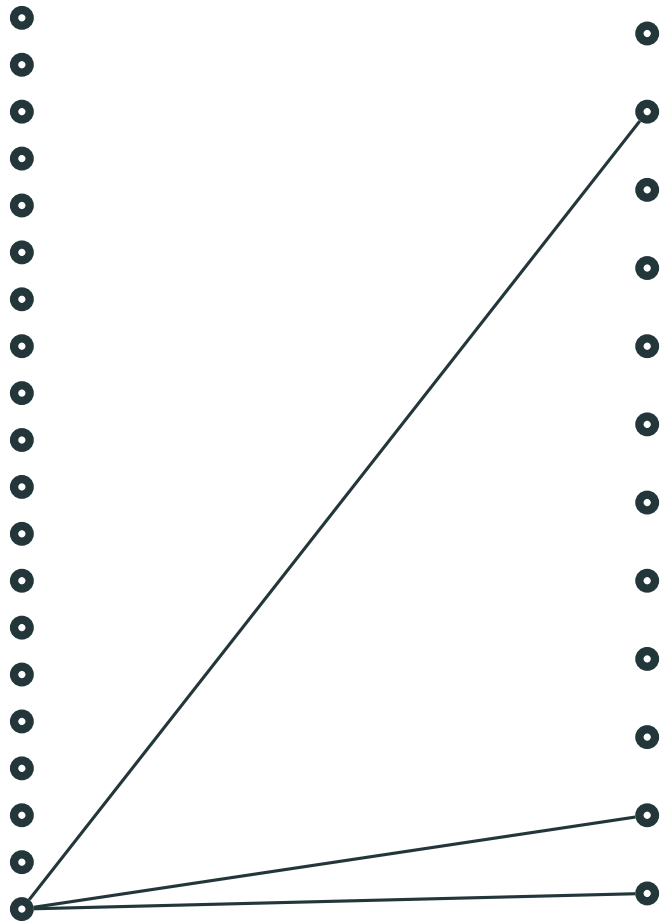




k problems

In Graph Terms

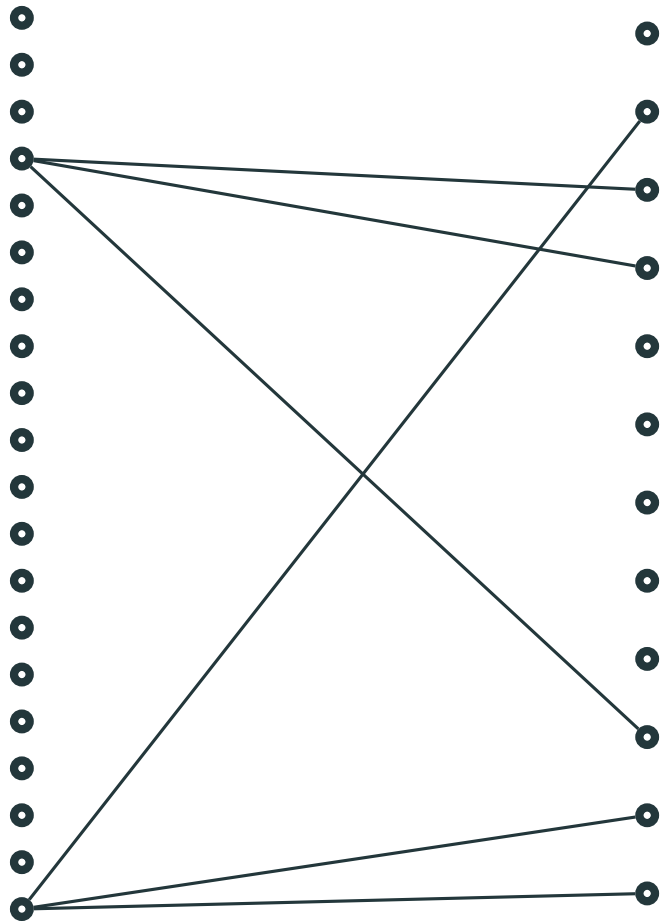
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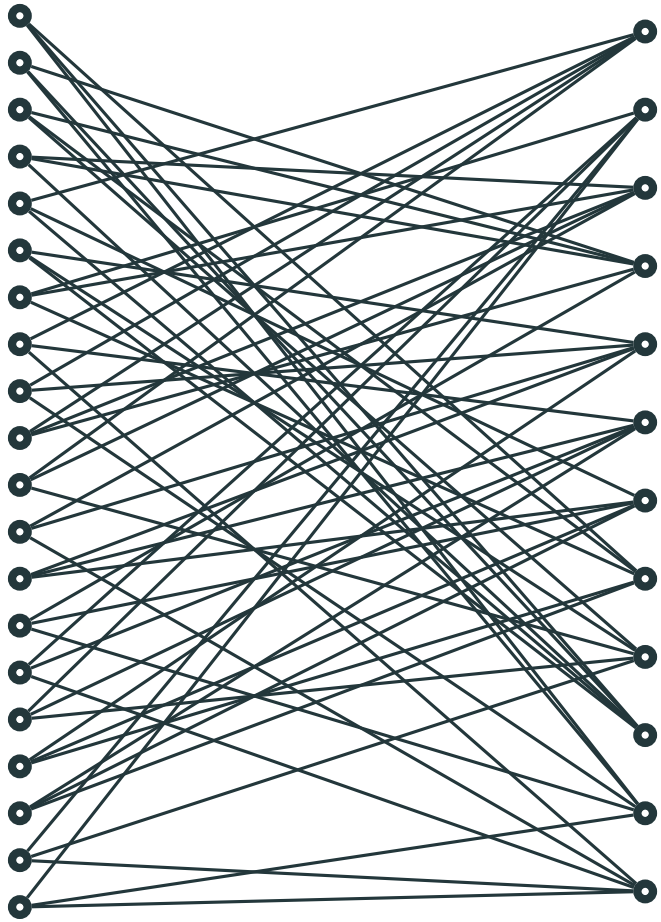
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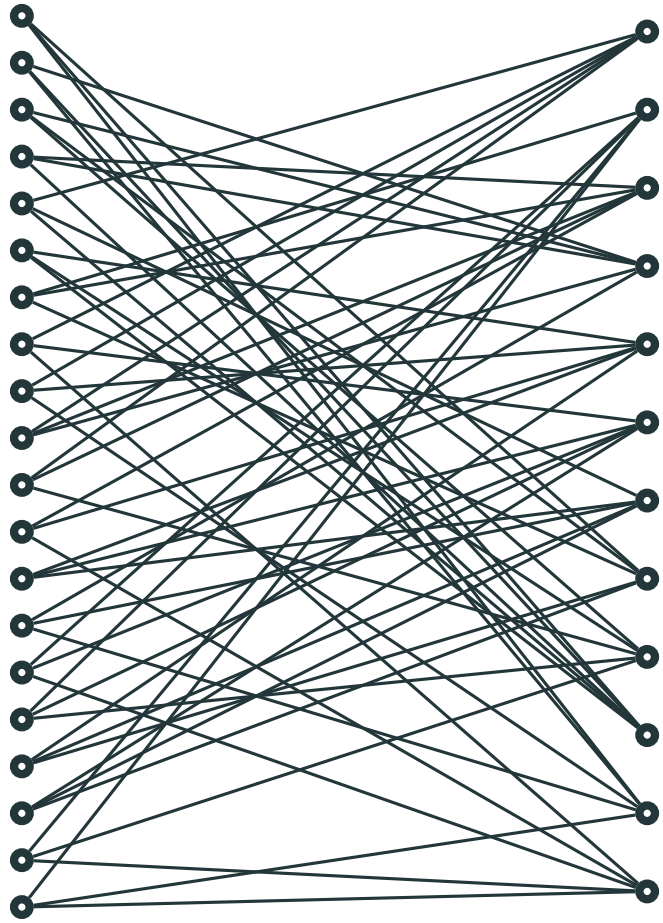
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In Graph Terms

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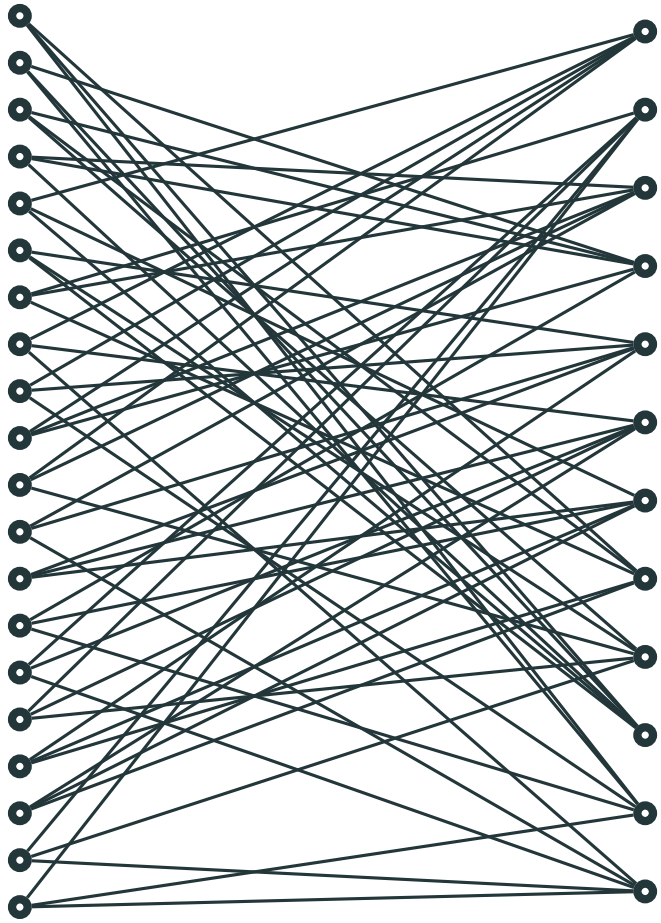


k problems

$$20 \times 3 = |E| = 5 \times k$$

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20 students



k problems

$k = 12$

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    - on one hand, the number of edges is equal to the total degree of the left part (i.e., the sum of all degree of the nodes from the left

Summary

- This is a bipartite graph: each edge joins a student and a problem
- We used the double counting technique to solve the problem:
    - on one hand, the number of edges is equal to the total degree of the left part (i.e., the sum of all degree of the nodes from the left
    - on the other hand, it is equal to the total degree of the right part